

Compact CNNs for Chest X-ray Screening: Accuracy, Explainability, and Containerised Deployment

Anonymous Submission

Anonymous Affiliation

Abstract

Lightweight convolutional networks are widely promoted for medical-image triage on resource-constrained hardware, on the assumption that smaller models translate into faster, cheaper deployment. We test this assumption empirically on a unified three-class chest X-ray task (Normal, Pneumonia, Tuberculosis) built from CheXpert (v1.0-small) and TBX11K with content-hash deduplication. A custom ~ 6.6 k-parameter CNN (*Victorio*) is compared against ImageNet-pretrained MobileNetV2, EfficientNet-B0 and ResNet-50 under two training regimes on a held-out test set of 1,353 images. The transfer backbones converge to 99.78–99.93% accuracy while *Victorio* reaches 92.4%, showing that capacity rather than pretraining domain drives performance. A Grad-CAM versus LIME Intersection-over-Union Agreement Index is tightly clustered at 0.534–0.537 across all eight model–regime pairs, indicating that the two post-hoc explainers converge on similar regions regardless of capacity. A full deployment benchmark on a Minikube cluster (5,040 requests across both CPU and GPU pods) exposes an architecture-dependent CPU/GPU crossover: GPU served ResNet-50 at $2.1\times$ the CPU median and EfficientNet-B0 at roughly $1.5\times$, tied with CPU on MobileNetV2, but CPU served the ultra-compact custom CNN $2.5\times$ faster than GPU. Tail latency on CPU reached up to $3\times$ its median, whereas GPU stayed within roughly $2\times$. The conventional pairing of a lightweight model with GPU acceleration is therefore inverted only at the lower bound of capacity; for ResNet-class backbones, GPU serving remains the right choice.

Keywords: Chest X-ray, CNN, Transfer learning, Explainability, Kubernetes deployment.

1 Introduction

Chest radiography remains the most common front-line imaging modality for respiratory disease, and convolutional neural networks (CNNs) routinely match or exceed expert performance on curated benchmarks [Rajpurkar et al., 2017]. Lightweight architectures such as MobileNetV2 [Sandler et al., 2018] and EfficientNet [Tan and Le, 2019] are increasingly favoured for clinical deployment, with the implicit claim that fewer parameters translate into faster, cheaper inference. In practice, end-to-end response time in a containerised service depends on preprocessing, HTTP, orchestration and model compute combined; the parameter-count advantage of a minimal network can be absorbed by the surrounding stack.

This paper presents a controlled empirical comparison addressing two hypotheses: **(H1)** that an ultra-compact custom CNN matches transfer-learning baselines on a balanced three-class CXR task; and **(H2)** that the CPU/GPU latency advantage is architecture-dependent, with GPU acceleration amortising its per-request launch overhead on heavier backbones but losing that advantage on compact models served at concurrency one. We make four contributions:

- A unified three-class CXR benchmark fusing CheXpert (Pneumonia) and TBX11K (Normal, Tuberculosis), with content-hash deduplication and stratified 80/10/10 splits.
- A capacity comparison of a custom CNN (~ 6.6 k parameters) against MobileNetV2, EfficientNet-B0 and ResNet-50 under ImageNet and CXR-from-scratch training regimes, on a held-out test set of 1,353 images.

- A quantitative explainability check using an Intersection-over-Union Agreement Index between Grad-CAM and LIME masks across all eight model–regime pairs.
- A full latency characterisation of the deployed inference service on a local Kubernetes (Minikube) cluster (5,040 requests, both CPU and GPU pods), showing that the CPU/GPU advantage flips with model size: GPU wins for ResNet-50 and EfficientNet-B0, ties for MobileNetV2, and loses to CPU only for the ultra-compact custom CNN.

2 Related Work

CNNs for chest radiography. Modern CXR pipelines almost universally rely on CNN backbones pre-trained on ImageNet, with reported AUROC values approaching radiologist performance for pneumonia and tuberculosis screening [Rajpurkar et al., 2017, Irvin et al., 2019, Liu et al., 2020]. Public datasets such as CheXpert [Irvin et al., 2019] and TBX11K [Liu et al., 2020] are typically studied in isolation; cross-dataset fusion introduces source-class confounds that must be acknowledged.

Lightweight architectures. Inverted residuals [Sandler et al., 2018] and compound scaling [Tan and Le, 2019] reduce parameter count by an order of magnitude while preserving accuracy on natural images. Whether their efficiency advantage transfers to deployed medical-imaging services has received less attention; reported speed improvements often refer to raw model FLOPs rather than to end-to-end response times in a realistic serving environment.

Explainability. Grad-CAM [Selvaraju et al., 2017] and LIME [Ribeiro et al., 2016] are the two most common post-hoc attribution methods used in radiology research. Quantitative comparisons between them are rare; pairwise agreement is typically reported only qualitatively, despite its importance as a sanity check against shortcut learning.

Containerised model serving. DNN serving systems such as Clockwork [Gujarati et al., 2020] target predictable per-request latency under containerised deployment, but report aggregate behaviour without decomposing it by model architecture and backend at the granularity relevant to single-image medical inference. We address this gap with a controlled per-architecture measurement on a Kubernetes cluster.

3 Materials and Methods

3.1 Dataset construction

A unified three-class corpus was assembled from two sources: the Pneumonia positives from CheXpert-Small [Irvin et al., 2019] (frontal, $n = 4,675$) and the Normal ($n = 4,178$) and Tuberculosis ($n = 4,798$) classes from TBX11K [Liu et al., 2020]. The pool was deduplicated using a content hash (SHA-256 of a 64×64 grayscale downsample, exact-match), removing 130 near-identical duplicates and yielding 13,521 unique studies. Stratified random splits produced training (10,816), validation (1,352) and test (1,353) subsets, with class balance preserved within ± 0.4 pp. Normal and TB originate from a single source (TBX11K) and Pneumonia from a single source (CheXpert); this source-class confound is addressed in Section 6.

3.2 Architectures

Four classifiers were compared. **MobileNetV2** [Sandler et al., 2018] (≈ 3.5 M parameters), **EfficientNet-B0** [Tan and Le, 2019] (≈ 5.3 M) and **ResNet-50** [He et al., 2016] (≈ 25 M) were initialised with ImageNet weights, with the final classification head replaced by a 3-way linear layer. **Victorio** is a custom minimal CNN of $\approx 6,627$ parameters: two 3×3 convolutional blocks ($3 \rightarrow 16 \rightarrow 32$ channels), each followed by ReLU

and 2×2 max-pooling, an adaptive average pooling to 4×4 , and a 3-way linear head. No batch normalisation or dropout. The design intentionally explores the lower bound of CNN capacity for this task.

3.3 Training regimes

Each architecture was trained under two regimes. The **ImageNet** regime initialises with ImageNet weights and uses ImageNet normalisation (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). The **CXR** regime initialises from scratch and uses dataset-specific normalisation (per-channel mean ≈ 0.534 , std ≈ 0.266 , computed from the training split). Images are single-channel grayscale replicated across three RGB channels. The two regimes therefore differ in both initialisation and normalisation, yielding eight model–regime pairs.

4 Experimental Setup

4.1 Training protocol

All models were trained with identical hyperparameters: input 256×256 , batch size 128, 60 epochs, AdamW ($lr = 10^{-4}$, weight decay 10^{-4}), cosine schedule with 3-epoch warm-up, label smoothing 0.05, mixed precision, early stopping (patience 8). Augmentation was random horizontal flip only. Resizing used aspect-ratio-preserving letterbox padding to avoid clipping costophrenic angles. All trainable parameters were updated end-to-end; no layers were frozen. Best-validation checkpoints were retained.

4.2 Evaluation and explainability

Accuracy, macro-averaged precision, recall and F1, and one-vs-rest macro AUROC were computed on the held-out test set ($n = 1,353$). Grad-CAM [Selvaraju et al., 2017] maps were extracted from the last convolutional block of each architecture; LIME [Ribeiro et al., 2016] explanations used 1,000 perturbations per image with quickshift segmentation. Both maps were thresholded at the 50th percentile and resized to 256×256 . The Agreement Index is the IoU of the two binary masks, summarised as mean $\pm$ standard deviation across one representative image per class for each model–regime pair.

4.3 Deployment protocol

Each best-validation checkpoint was packaged in a containerised FastAPI service with a single `/predict` POST endpoint, with model selection exposed as a query parameter so all eight checkpoints were served from one running pod. CPU and GPU images were built with Podman in rootless mode using fully offline wheel installation; the CPU image is based on `python:3.10-slim` and the GPU image on `nvidia/cuda:12.8.0-runtime-ubuntu22.04`. Both images were deployed as separate Pods on a local Minikube cluster (Docker driver, 6 vCPU, 16 GB RAM; the GPU pod scheduled on a host-attached NVIDIA RTX 5090 exposed via the NVIDIA device plugin). Services were exposed via `kubectl port-forward`. For each (backend, model, image) triple, 5 warm-up requests followed by 100 measured requests were issued sequentially using the Python `requests` library; latency was recorded client-side as the wall-clock time of the HTTP round trip. The full data set totals 5,040 requests with no failures, in line with established service-latency reporting practice [Dean and Barroso, 2013].

5 Results

5.1 Classification performance

Table 1 reports test-set metrics for all eight model–regime pairs. The three transfer-learning backbones cluster at 99.78–99.93% accuracy, with F1 and AUROC indistinguishable to two decimal places. Victorio reaches 92.4% under the CXR regime and 91.9% under the ImageNet regime, trailing the transfer baselines by roughly

7 percentage points. The training-regime choice has only a marginal effect (a 0.5–1.5 pp swing depending on architecture) and no consistent direction. Victorio’s residual errors fall almost entirely on the TB/Normal boundary, with Pneumonia consistently well-separated.

Model	Regime	Acc.	Prec.	Rec.	F1	AUROC
EfficientNet-B0	CXR	0.9985	0.9985	0.9985	0.9985	1.0000
EfficientNet-B0	ImageNet	0.9985	0.9985	0.9985	0.9985	0.9999
MobileNetV2	CXR	0.9978	0.9976	0.9979	0.9977	1.0000
MobileNetV2	ImageNet	0.9993	0.9992	0.9993	0.9992	0.9993
ResNet-50	CXR	0.9993	0.9992	0.9993	0.9992	0.9997
ResNet-50	ImageNet	0.9993	0.9992	0.9993	0.9992	1.0000
Victorio	CXR	0.9239	0.9226	0.9235	0.9227	0.9818
Victorio	ImageNet	0.9187	0.9173	0.9181	0.9174	0.9807

Table 1: Test-set performance ($n = 1,353$) for all model×regime pairs. Macro-averaged precision/recall/F1; one-vs-rest macro AUROC.

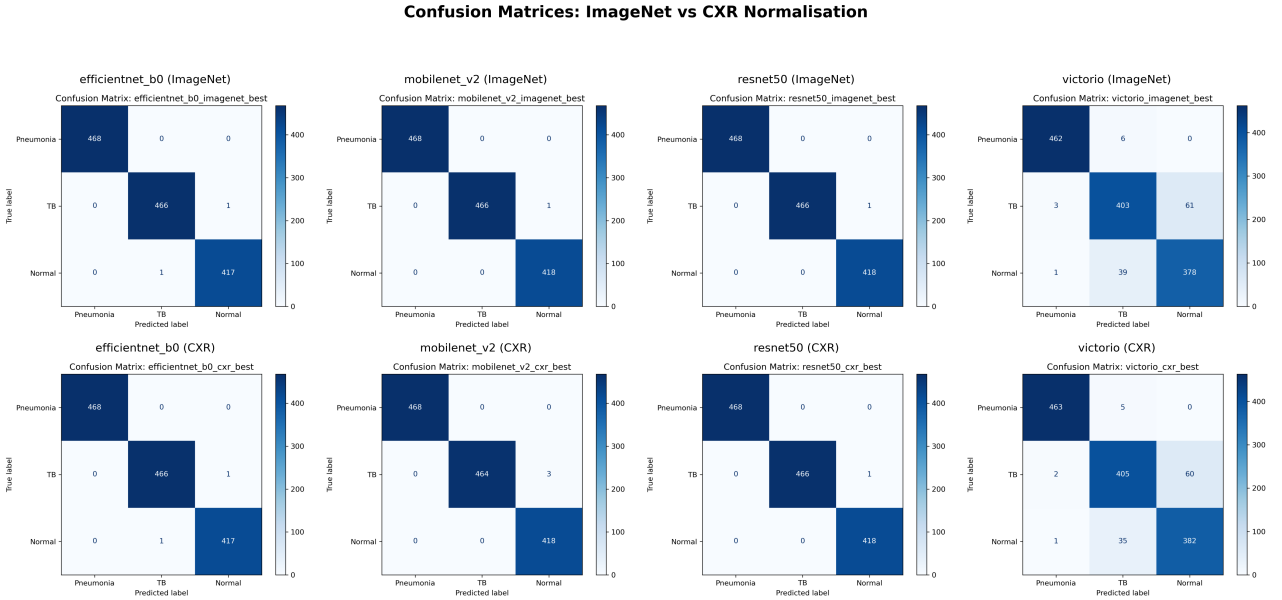


Figure 1: Confusion matrices for all eight model–regime pairs on the held-out test set ($n = 1,353$). Top row: ImageNet regime; bottom row: CXR regime. Columns (left to right): EfficientNet-B0, MobileNetV2, ResNet-50, Victorio. The three transfer backbones produce near-perfect diagonals; Figure 1 shows the per-class breakdown: Victorio’s residual errors fall almost entirely on the TB→Normal boundary, with Pneumonia consistently well-separated.

5.2 Explainability and Agreement Index

The narrow variance in Grad-CAM versus LIME agreement across the eight model–regime pairs suggests that explainer consistency is governed more by image-level pathology localisation than by representational depth: the discriminative region of a CXR image is the same target for a 6,600-parameter network and a 25 M-parameter network, and gradient-based and perturbation-based methods both find it. Figure 2 reports the Agreement Index numerically: mean IoU values lie within a 0.003-wide band (0.534–0.537) with standard deviations near 0.40, confirming that the two post-hoc explainers converge on similar regions regardless of architecture or training regime. The global mean of ≈ 0.535 also hides a clean per-class pattern: Pneumonia

cases produce near-perfect Grad-CAM versus LIME overlap, while Normal and TB cases yield substantially lower IoU, consistent with the Pneumonia opacity being a larger and more spatially coherent target than the focal patterns typical of TB. Qualitatively (see Figure 4 in Appendix A), the three transfer backbones produce heatmaps tightly centred on the affected lung region, while Victorio’s Grad-CAM covers the correct lung area in a broader, less sharply defined pattern, consistent with its lower capacity. Across the examined examples, neither method showed dominant activation on non-anatomical artefacts (borders, text markers, image corners), reducing concerns about shortcut learning from dataset-specific cues.

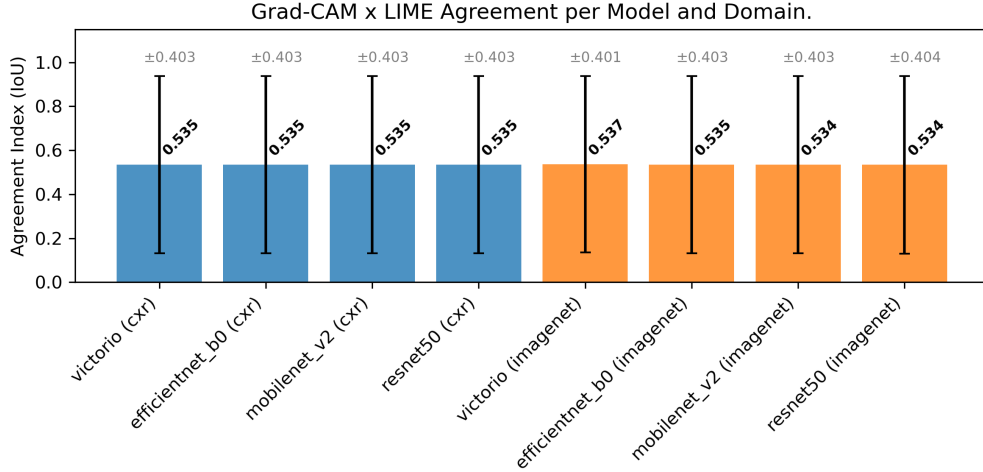


Figure 2: Grad-CAM versus LIME Agreement Index (IoU of binary masks) across the eight model–regime pairs. Mean values are tightly clustered (0.534–0.537); standard deviations near 0.40 reflect per-image variation typical of gradient-based versus perturbation-based attributions.

5.3 Deployment latency

Table 2 reports warm-state per-request latency for all eight model–regime pairs on both backends, aggregated over 100 measured requests per (backend, model, image) cell across three representative test images (one per class). The first 5 requests of each cell were treated as warm-up and excluded from the warm statistics. Predicted classes were identical across CPU and GPU for every cell, confirming functional equivalence of the deployed models across devices.

Model	Regime	CPU pod		GPU pod	
		P50 (ms)	P95 (ms)	P50 (ms)	P95 (ms)
MobileNetV2	CXR	101	924	93	262
MobileNetV2	ImageNet	69	507	93	273
EfficientNet-B0	CXR	161	1010	110	300
EfficientNet-B0	ImageNet	188	1159	112	301
ResNet-50	CXR	517	1430	250	432
ResNet-50	ImageNet	398	1445	248	431
Victorio	CXR	19	113	48	204
Victorio	ImageNet	19	111	47	210

Table 2: Warm-state per-request latency (300 measured requests per row, aggregated across three test images). Values are median (P50) and 95th percentile (P95) in milliseconds. Lower is better.

Figure 3 shows that the CPU-versus-GPU comparison is architecture-dependent. GPU halved the median latency of ResNet-50 (250 ms vs 517 ms on the CXR regime, a $2.1\times$ speed-up) and gave a smaller but consistent

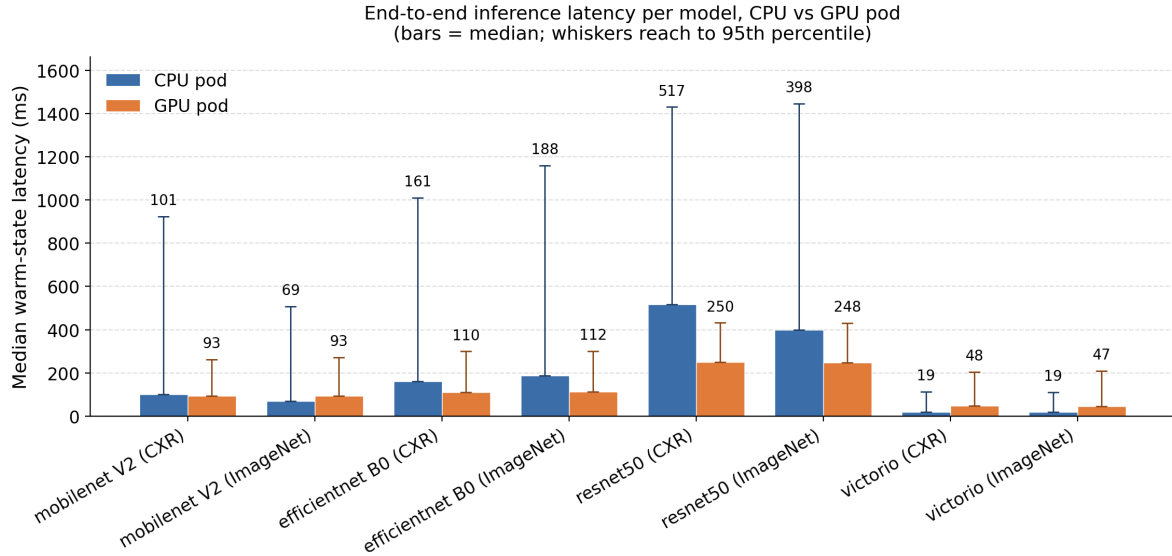


Figure 3: Warm-state median latency per model, CPU pod versus GPU pod (whiskers reach to the 95th percentile across 300 measured requests per cell). The CPU/GPU advantage flips with model size: GPU wins for ResNet-50 and EfficientNet-B0, ties for MobileNetV2, and CPU wins only for the ultra-compact custom CNN.

advantage on EfficientNet-B0 (110 ms vs 161 ms, roughly 1.5 $\times$). For MobileNetV2 the two backends were within tens of milliseconds and effectively tied. Only for the ultra-compact custom CNN did CPU clearly win, serving at 19 ms against 48 ms on GPU, a 2.5 $\times$ inversion. Tail behaviour also diverged: CPU P95 reached up to three times its own median (1,430 ms vs 517 ms for ResNet-50 CXR), whereas GPU P95 stayed within roughly twice the median across all models.

6 Discussion

Capacity, not training regime, drives accuracy. The transfer backbones converge within a ~ 0.15 pp band, and the ImageNet vs. CXR choice has at most a 1.5 pp effect on Victorio and under 1 pp for the larger models. Hypothesis H1 is therefore not supported: an ultra-compact custom CNN does not match transfer-learning architectures on this task. The practical ~ 7 pp gap is only relevant where the deployment target genuinely cannot host a 5–25 M parameter model, which is rarely the case in modern hospital infrastructure.

Source-class confound. We fused CheXpert (Pneumonia) and TBX11K (Normal, TB) to obtain a balanced three-class corpus, but the classes are not source-balanced; a model could in principle classify the dataset of origin rather than the underlying condition [Zech et al., 2018]. Two pieces of evidence mitigate this: Victorio reaches only 92% accuracy despite trivial access to the same source cues, and neither Grad-CAM nor LIME shows dominant activation on non-anatomical artefacts. We flag this as the principal limitation and recommend leave-one-source-out and adversarial-domain validation in future work.

Explainer agreement is invariant to capacity. The Agreement Index range of 0.534–0.537 across all model–regime pairs indicates that gradient-based and perturbation-based explainers converge on the same regions regardless of whether the underlying model has 6,600 or 25 M parameters. The absolute level of ≈ 0.535 sits in the mid-range where two post-hoc explainers agree on the coarse anatomical region but diverge on its precise boundary, a typical signature of gradient-based versus perturbation-based attribution methods.

Backend choice is architecture-dependent (H2). H2 is partially supported: the conventional CPU-faster-for-small-models expectation holds only at the lower bound of capacity. GPU per-request overhead (kernel launch, host-to-device transfer of a 256×256 tensor, CUDA stream synchronisation) is roughly constant, so it is amortised by the heavier forward passes of ResNet-50 and EfficientNet-B0 but dominates the compute on the ultra-compact custom CNN, inverting the ordering. For low-throughput single-image CXR screening, the practical guidance is therefore split: pair compact backbones with CPU pods, but retain GPU serving for ResNet-class architectures, where it halves median latency. The CPU pod also showed a heavy right tail (P95 up to $3 \times$ its median), so GPU serving is preferable when worst-case rather than typical latency is the operational constraint.

Limitations. The benchmark was conducted at concurrency level one on a single-node Minikube cluster; throughput-bound or concurrent workloads were not studied and would likely shift the CPU/GPU balance. CheXpert and TBX11K were used at native frontal-only resolutions, and we did not investigate cross-institutional generalisation. Clinical translation would additionally require validation under regulatory frameworks (e.g. EU MDR, GDPR), which lies beyond the scope of this study.

7 Conclusion

We compared a $\sim 6.6\text{k}$ -parameter custom CNN with three ImageNet-pretrained backbones on a deduplicated three-class CXR corpus, and benchmarked all four models on a containerised FastAPI service in a local Minikube cluster. Three findings stand out: a ~ 7 pp capacity gap consistent across training regimes; explainer agreement invariant to capacity at ≈ 0.535 ; and a deployment benchmark (5,040 requests, eight model-regime pairs, both backends) showing an architecture-dependent crossover (GPU $2.1 \times$ faster than CPU on ResNet-50, $\sim 1.5 \times$ on EfficientNet-B0, tied on MobileNetV2, $2.5 \times$ slower than CPU on the ultra-compact custom CNN). For low-throughput single-image CXR screening on commodity hardware, compact backbones should be paired with CPU pods, while GPU serving remains the right choice for heavier architectures.

Acknowledgments

A Grad-CAM and LIME on a Pneumonia case

Figure 4 shows Grad-CAM heatmaps and LIME superpixel masks for a representative Pneumonia test image across the four architectures and two training regimes, providing qualitative support for the Agreement Index reported in Section 5.

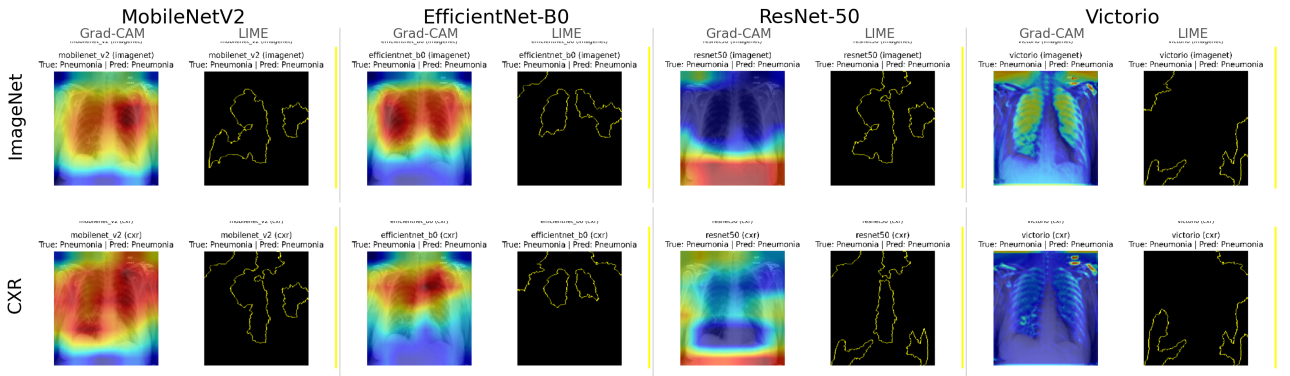


Figure 4: Grad-CAM and LIME maps on a representative Pneumonia test image. Top row: ImageNet regime; bottom row: CXR regime. Columns: MobileNetV2, EfficientNet-B0, ResNet-50, Victorio. Each cell shows the Grad-CAM heatmap (left) and the LIME superpixel mask (right) for the same case.

References

- [Dean and Barroso, 2013] Dean, J. and Barroso, L. A. (2013). The tail at scale. *Communications of the ACM*, 56(2):74–80.
- [Gujarati et al., 2020] Gujarati, A., Karimi, R., Alzayat, S., Hao, W., Kaufmann, A., Vigfusson, Y., and Mace, J. (2020). Serving DNNs like clockwork: Performance predictability from the bottom up. In *Proceedings of the 14th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, pages 443–462.
- [He et al., 2016] He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.
- [Irvin et al., 2019] Irvin, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Ilcus, S., Chute, C., Marklund, H., Haghighi, B., Ball, R., Shpanskaya, K., et al. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 590–597.
- [Liu et al., 2020] Liu, Y., Wu, Y.-H., Ban, Y., Wang, H., and Cheng, M.-M. (2020). Rethinking computer-aided tuberculosis diagnosis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2646–2655.
- [Rajpurkar et al., 2017] Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., and Ng, A. Y. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv preprint arXiv:1711.05225*.
- [Ribeiro et al., 2016] Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). “Why should I trust you?”: Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 1135–1144.
- [Sandler et al., 2018] Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., and Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520.
- [Selvaraju et al., 2017] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 618–626.
- [Tan and Le, 2019] Tan, M. and Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 6105–6114.
- [Zech et al., 2018] Zech, J. R., Badgeley, M. A., Liu, M., Costa, A. B., Titano, J. J., and Oermann, E. K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11):e1002683.